

The root system of the Lie algebra $\mathfrak{sl}_3(\mathbb{C})$

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Overview

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- 2 Root system of $\mathfrak{sl}_3(\mathbb{C})$

Definition

The special linear Lie algebra of order 3 (denoted $\mathfrak{sl}_3(\mathbb{C})$) is the Lie algebra of 3×3 complex matrices with trace zero and with the Lie bracket $[X, Y] := XY - YX$.

Ex.

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & -a-e \end{pmatrix} \in \mathfrak{sl}_3(\mathbb{C})$$

Lie algebra $\mathfrak{sl}_3(\mathbb{C})$

Basis of $\mathfrak{sl}_3(\mathbb{C})$

$$H_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad X_1 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

$$X_2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad X_3 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

$$H_2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad Y_1 = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

$$Y_2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad Y_3 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}.$$

$$\begin{aligned}[H_1, X_1] &= H_1 X_1 - X_1 H_1 \\&= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \\&= \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\&= \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = X_1\end{aligned}$$

Lie algebra $\mathfrak{sl}_3(\mathbb{C})$

Table: Lie bracket multiplication table

$[\cdot, \cdot]$	H_1	X_1	X_2	X_3	H_2	Y_1	Y_2	Y_3
H_1	0	X_1	X_2	$2X_3$	0	$-Y_1$	$-Y_2$	$-2Y_3$
X_1	$-X_1$	0	X_3	0	X_1	$H_1 - H_2$	0	$-Y_2$
X_2	$-X_2$	$-X_3$	0	0	$-2X_2$	0	H_2	Y_1
X_3	$-2X_3$	0	0	0	$-X_3$	$-X_2$	X_1	H_1
H_2	0	$-X_1$	$2X_2$	X_3	0	Y_1	$-2Y_2$	$-Y_3$
Y_1	Y_1	$H_2 - H_1$	0	X_2	$-Y_1$	0	$-Y_3$	0
Y_2	Y_2	0	$-H_2$	$-X_1$	$2Y_2$	Y_3	0	0
Y_3	$2Y_3$	Y_2	$-Y_1$	$-H_1$	Y_3	0	0	0

Definition

$$\begin{aligned}\mathfrak{h} &:= \text{span}_{\mathbb{C}} \left\{ H_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}, H_2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \right\} \\ &= \left\{ \begin{pmatrix} a & 0 & 0 \\ 0 & e & 0 \\ 0 & 0 & -a-e \end{pmatrix} : a, e \in \mathbb{C} \right\}\end{aligned}$$

Lie algebra $\mathfrak{sl}_3(\mathbb{C})$

$[\cdot, \cdot]$	H_1	X_1	X_2	X_3	H_2	Y_1	Y_2	Y_3
H_1	0	X_1	X_2	$2X_3$	0	$-Y_1$	$-Y_2$	$-2Y_3$
X_1	$-X_1$	0	X_3	0	X_1	$H_1 - H_2$	0	$-Y_2$
X_2	$-X_2$	$-X_3$	0	0	$-2X_2$	0	H_2	Y_1
X_3	$-2X_3$	0	0	0	$-X_3$	$-X_2$	X_1	H_1
H_2	0	$-X_1$	$2X_2$	X_3	0	Y_1	$-2Y_2$	$-Y_3$
Y_1	Y_1	$H_2 - H_1$	0	X_2	$-Y_1$	0	$-Y_3$	0
Y_2	Y_2	0	$-H_2$	$-X_1$	$2Y_2$	Y_3	0	0
Y_3	$2Y_3$	Y_2	$-Y_1$	$-H_1$	Y_3	0	0	0

Lie algebra $\mathfrak{sl}_3(\mathbb{C})$

$[\cdot, \cdot]$	H_1	X_1	X_2	X_3	H_2	Y_1	Y_2	Y_3
H_1	0	X_1	X_2	$2X_3$	0	$-Y_1$	$-Y_2$	$-2Y_3$
X_1	$-X_1$	0	X_3	0	X_1	$H_1 - H_2$	0	$-Y_2$
X_2	$-X_2$	$-X_3$	0	0	$-2X_2$	0	H_2	Y_1
X_3	$-2X_3$	0	0	0	$-X_3$	$-X_2$	X_1	H_1
H_2	0	$-X_1$	$2X_2$	X_3	0	Y_1	$-2Y_2$	$-Y_3$
Y_1	Y_1	$H_2 - H_1$	0	X_2	$-Y_1$	0	$-Y_3$	0
Y_2	Y_2	0	$-H_2$	$-X_1$	$2Y_2$	Y_3	0	0
Y_3	$2Y_3$	Y_2	$-Y_1$	$-H_1$	Y_3	0	0	0

Lie algebra $\mathfrak{sl}_3(\mathbb{C})$

$$\begin{aligned}[H_1, X_1] &= 1 \cdot X_1, & [H_2, X_1] &= -1 \cdot X_1; \\ [H_1, X_2] &= 1 \cdot X_2, & [H_2, X_2] &= 2 \cdot X_2; \\ [H_1, X_3] &= 2 \cdot X_3, & [H_2, X_3] &= 1 \cdot X_3;\end{aligned}$$

$$\begin{aligned}[H_1, Y_1] &= -Y_1, & [H_2, Y_1] &= Y_1; \\ [H_1, Y_2] &= -Y_2, & [H_2, Y_2] &= -2Y_2; \\ [H_1, Y_3] &= -2Y_3, & [H_2, Y_3] &= -Y_3.\end{aligned}$$

Definition

Define $\beta_1, \beta_2, \beta_3 \in \mathfrak{h}^*$ as

$\beta_1 : \mathfrak{h} \rightarrow \mathbb{C}$	$\beta_2 : \mathfrak{h} \rightarrow \mathbb{C}$	$\beta_3 := (\beta_1 + \beta_2) : \mathfrak{h} \rightarrow \mathbb{C}$
$H_1 \mapsto 1$	$H_1 \mapsto 1$	$H_1 \mapsto 2$
$H_2 \mapsto -1$	$H_2 \mapsto 2$	$H_2 \mapsto 1$

$$\begin{aligned}[H_1, X_1] &= \beta_1(H_1)X_1, & [H_2, X_1] &= \beta_1(H_2)X_1; \\[H_1, X_2] &= \beta_2(H_1)X_2, & [H_2, X_2] &= \beta_2(H_2)X_2; \\[H_1, X_3] &= \beta_3(H_1)X_3, & [H_2, X_3] &= \beta_3(H_2)X_3;\end{aligned}$$

$$\begin{aligned}[H_1, Y_1] &= -\beta_1(H_1)Y_1, & [H_2, Y_1] &= -\beta_1(H_2)Y_1; \\[H_1, Y_2] &= -\beta_2(H_1)Y_2, & [H_2, Y_2] &= -\beta_2(H_2)Y_2; \\[H_1, Y_3] &= -\beta_3(H_1)Y_3, & [H_2, Y_3] &= -\beta_3(H_2)Y_3.\end{aligned}$$

Definition

A linear map $\text{ad}_X : \mathfrak{sl}_3(\mathbb{C}) \rightarrow \mathfrak{sl}_3(\mathbb{C})$ by

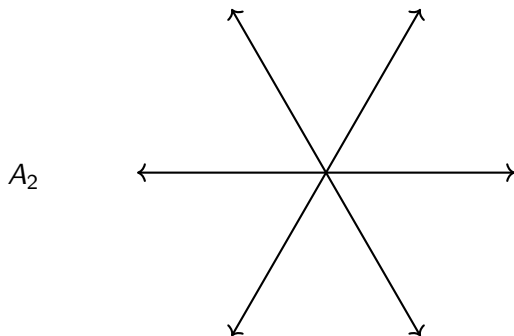
$$\text{ad}_X(Y) = [X, Y].$$

The map $X \mapsto \text{ad}_X$ is the **adjoint map**.

Root system of $\mathfrak{sl}_3(\mathbb{C})$

claim

$R = \{\pm\beta_1, \pm\beta_2, \pm\beta_3\}$ becomes a root system A_2



Root system of $\mathfrak{sl}_3(\mathbb{C})$

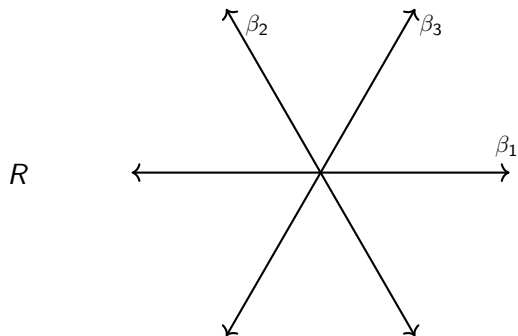
Definition

$$(\beta_i, \beta_j) := \text{trace}(\text{ad}_{\mathbf{H}_{\beta_i}} \circ \text{ad}_{\mathbf{H}_{\beta_j}})$$

Table: Inner product of $\mathfrak{h}^*$

$(\cdot, \cdot)$	β_1	$-\beta_1$	β_2	$-\beta_2$	β_3	$-\beta_3$
β_1	1/3	-1/3	-1/6	1/6	1/6	-1/6
$-\beta_1$		1/3	1/6	-1/6	-1/6	1/6
β_2			1/3	-1/3	1/6	-1/6
$-\beta_2$				1/3	-1/6	1/6
β_3					1/3	-1/3
$-\beta_3$						1/3

Root system of $\mathfrak{sl}_3(\mathbb{C})$



References



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